ResTier

Tiered Storage Engine for ResilientDB

Master's Project Presentation
Under the guidance of Prof M. Sadoghi

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ExpoLab
Creativity Unfolded

Apache ResilientDB
Incubating

UCDAVIS
COMPUTER SCIENCE

What is ResilientDB

ResilientDB is a high-throughput distributed ledger built on scale-centric design principles, aimed at democratizing and decentralizing computation. Originally conceived at UC Davis's ExpoLab, it is now an Apache Incubating project.

Key features -

- Renewed the vision of Bitcoin/blockchain - integrating privacy, integrity, transparency, and accountability into the computational model
- Built around fault-tolerant BFT consensus protocols (to ensure the system continues operating correctly even when some nodes act maliciously or fail arbitrarily), redesigned for modern performance demands
- Uses parallelism and deep pipelining at every layer, optimized for modern hardware and global cloud infrastructure

Why ResTier?

Present Scenario

- All data stays in local LevelDB forever (SSD bloat)
- Storage grows unboundedly as the blockchain accumulates history (append-only by design)
- No mechanism to offload cold/archival data
- Historical data is less frequently queried

Why ResTier?

Proposed Solution

- Use IPFS as a cold tier to migrate historical data asynchronously based on configurable thresholds and checkpoints

Why IPFS?

- Content-addressed (CID = cryptographic hash of data)
- Trustless verification of content
- Deduplication built-in -> same content = same CID
- Peer-to-peer -> replicates across nodes automatically
- No vendor lock-in -> open source protocol

Problem Statement

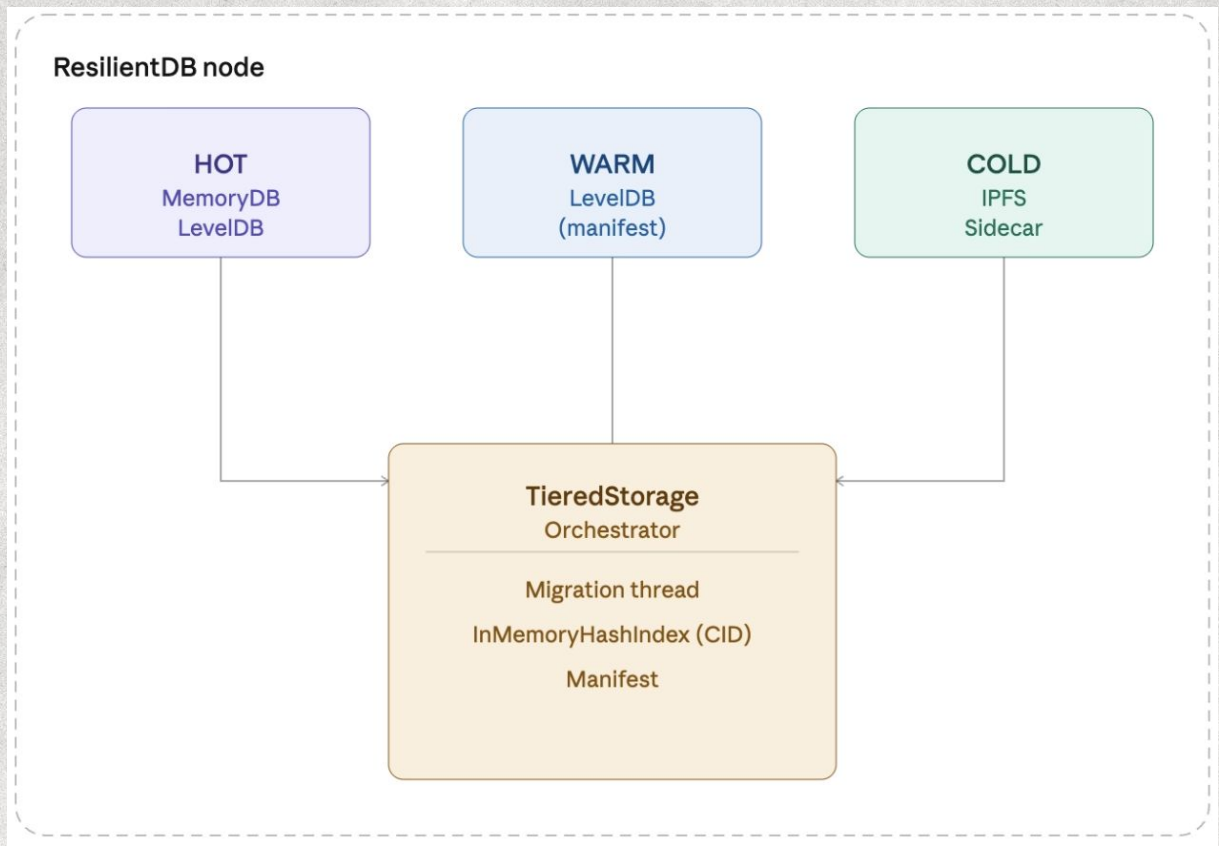
Design and implement a tiered storage engine for ResilientDB that:

1. Transparently migrates historical data from hot storage to IPFS
2. Maintains PBFT consensus latency (no write-path overhead)
3. Provides O(1) cold-read performance via an in-memory secondary index
4. Survives crashes with zero data loss
5. Operates in 4 storage modes to accommodate different deployment needs

Constraints:

- Each PBFT replica migrates independently (no cross-node consensus for migration)
- Zero data-loss window during migration
- Block cache must be invalidated on eviction (LevelDB hot tier)
- Minimal effect on write throughput and read latency

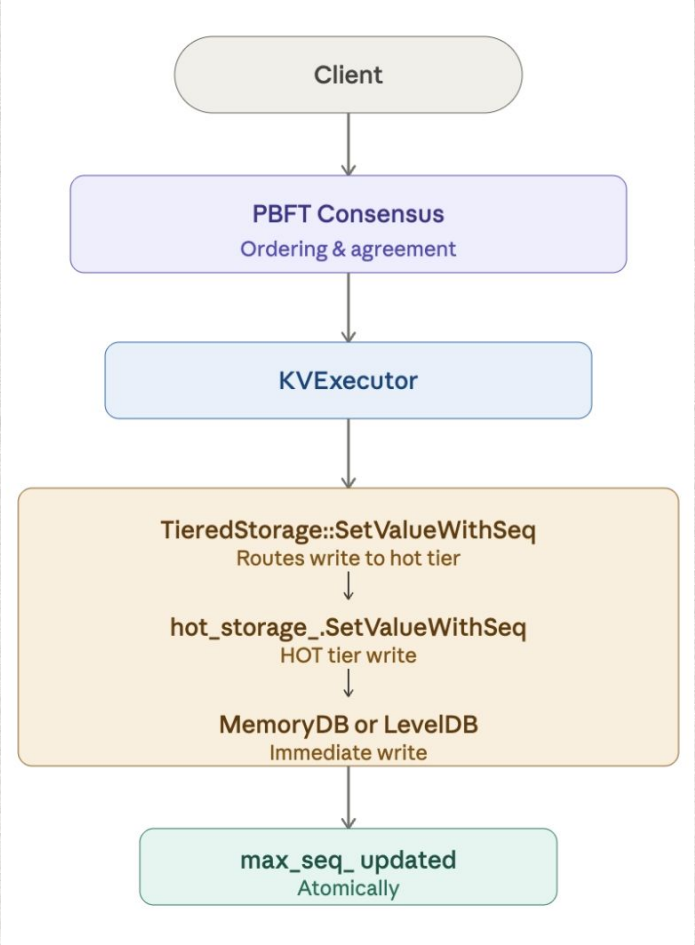
System Architecture



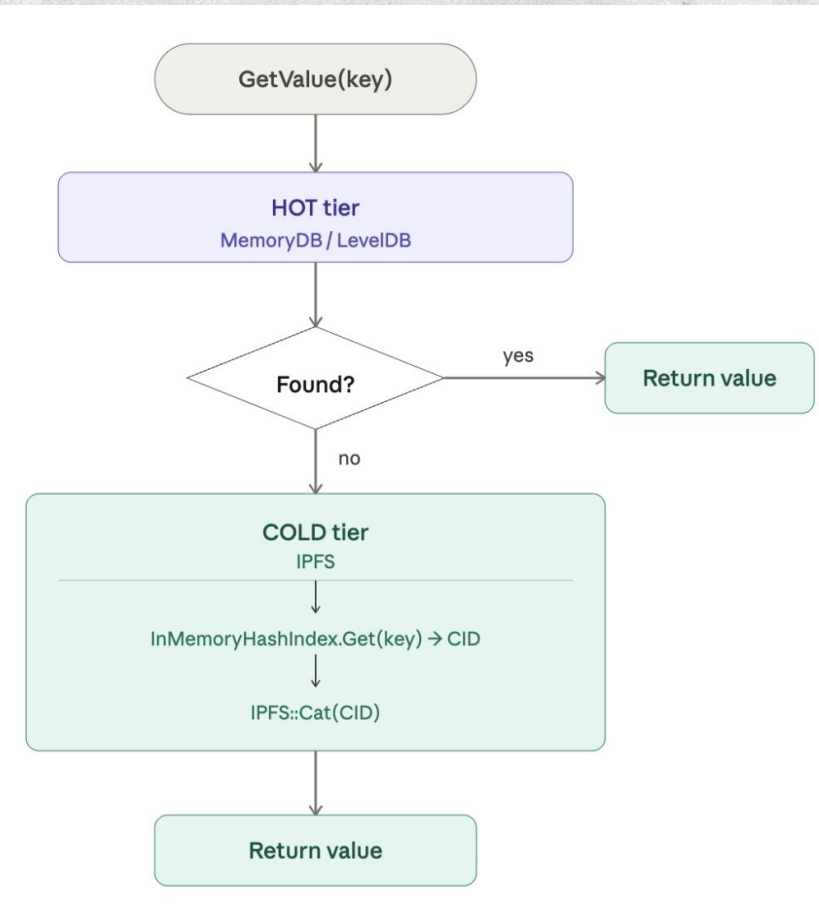
Four Storage Modes

Mode	Backend	Hot Tier	Warm Tier	Cold Tier	Use Case
0	MEMORYDB	MemoryDB	—	—	Dev/testing, fastest
1	LEVELDB	LevelDB	—	—	Production, small data
2	TIERED	LevelDB	LevelDB (manifest)	IPFS	Production, large data
3	TIERED	MemoryDB	LevelDB (manifest)	IPFS	High-throughput, crash-tolerant

Write Path (No Overhead)



Read Path (Fallback)



Crash Recovery

Four windows analyzed:

Window	State	GET behavior	Safe?
After IPFS upload, before index add	Data in IPFS + hot, NOT in index	Hits hot → correct	✅
After index add, before hot delete	Data in IPFS + index + hot	Hits hot → correct	✅
After hot delete	Data in IPFS + index only	Index→CID→IPFS Cat	✅
After hot delete, LRU cache stale	Stale entry in block cache	**Was: stale value. Fixed:** `Remove(key)`	✅

Zero data-loss. At worst, a key exists in both hot and cold (safe duplicate).

Conclusion

ResTier delivers:

- Transparent hot→cold data migration with zero write-path overhead
- O(1) cold-read performance via in-memory secondary index
- Safe concurrent reads during migration — 0 mismatches verified
- Crash recovery with no data loss
  4 storage modes for different deployment needs
- Works with existing PBFT consensus no consensus changes

Thank You

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